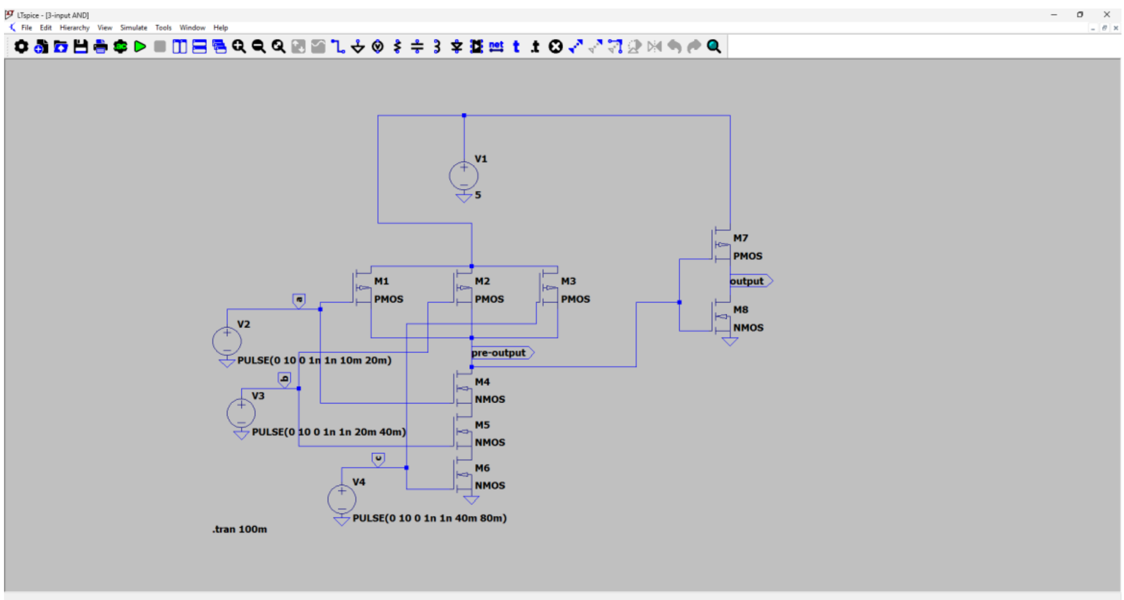
# MOSFET Logic Gates

**ENGS152 Circuits Lab**

**2 May 2026**

## Step 1: 3-input AND

### Schematic Circuit

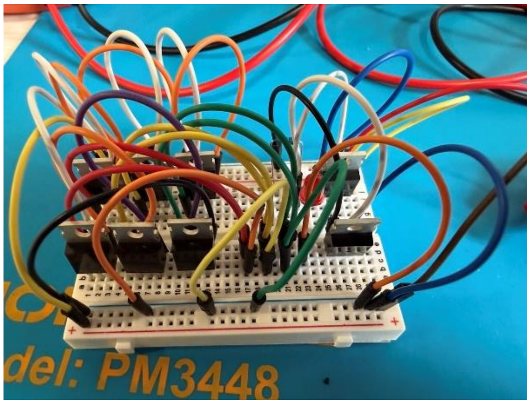


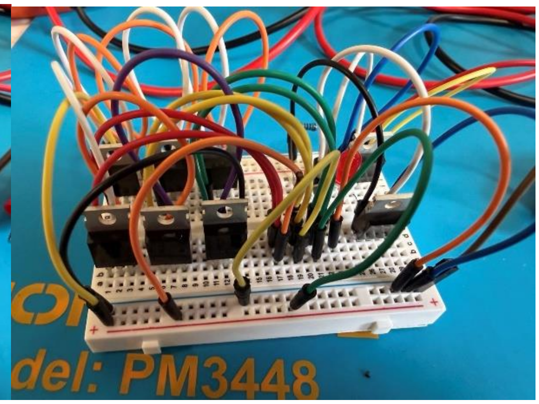

### Truth Table

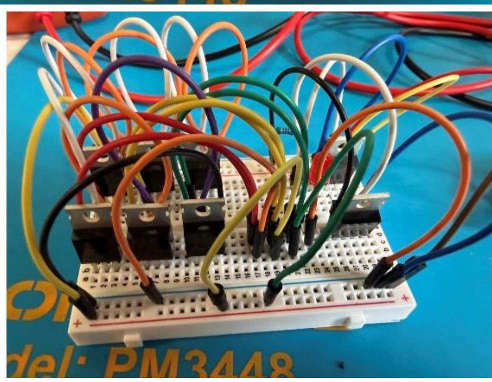
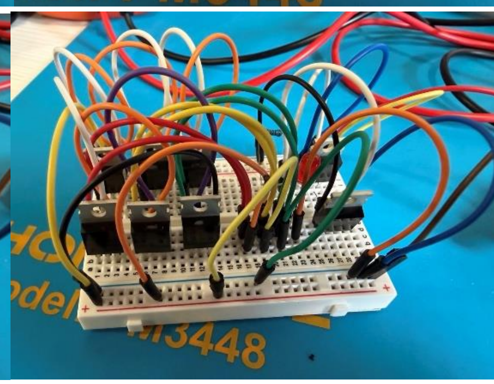
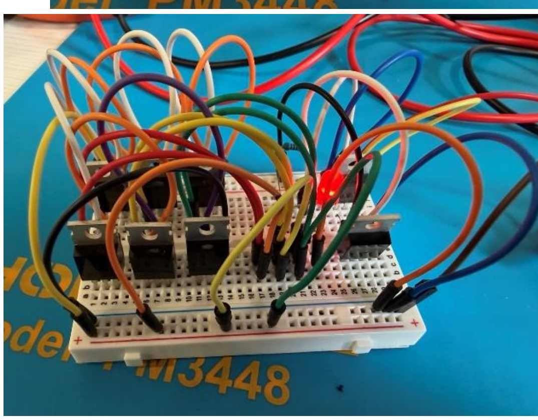
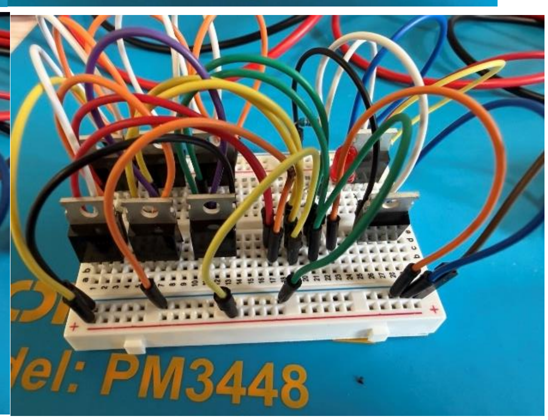
| A | B | C | X |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 |

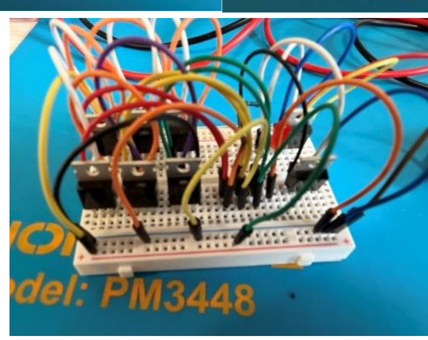
## Real-life assemblies with different input cases

Orange – Yellow – Green jumper wires as input pins



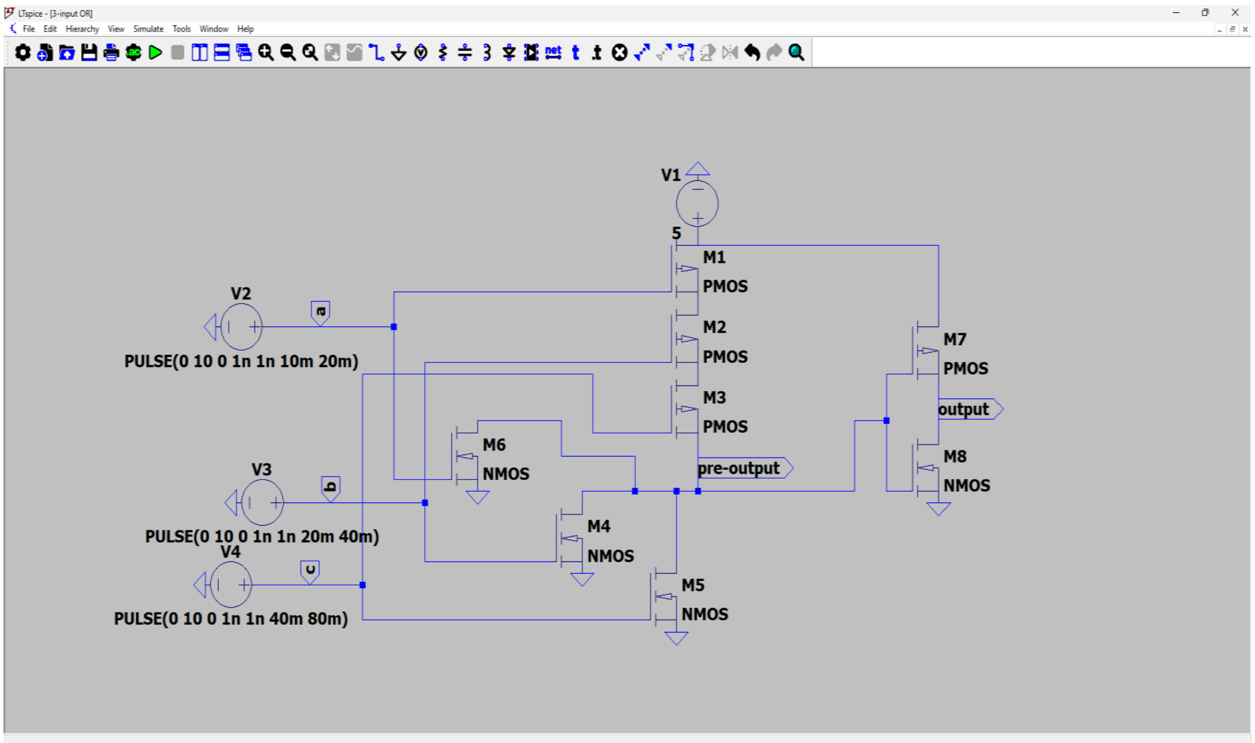




In this particular circuit PMOS are connected in parallel and NMOS is series. The circuit is 3-input NAND circuit connected to Inverter MOSFET circuit's gate. NAND gives logical output 0 only when all inputs are 1, and the inverter flips that 1 to 0, hence the output becomes 1 only when all 3 inputs are 1.

# Step 2: 3-input OR

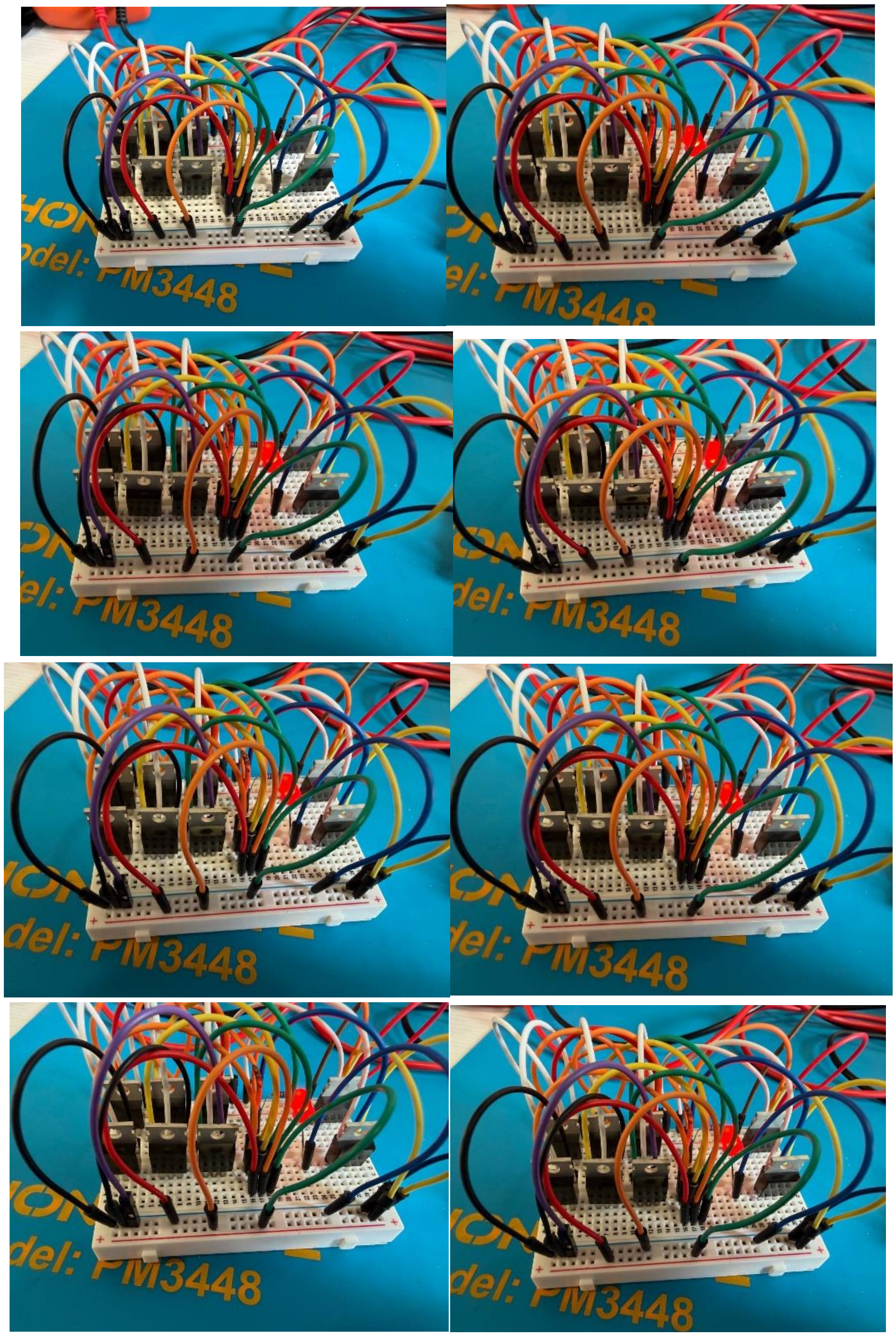



## Truth Table

| A | B | C | X |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 |

## Real-life assemblies with different input cases

Red – Yellow – Green jumper wires as input pins




In this particular circuit PMOS are connected in series and NMOS in parallel. The circuit is 3-input NOR circuit connected to Inverter MOSFET circuit's gate. NOR gives logical output 1 only when all inputs are 0, otherwise the output is always 1. The inverter flips that 1 to 0, and 0 to 1, hence the output becomes 0 only when all 3 inputs are 0, and 1 when at least one pin is 1.